

Student Document Security System

D.Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY,
KASABA BAWADA, KOLHAPUR
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(Academic Year: 2024-25)



D Y P A T I L
COLLEGE OF
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)
KASABA BAWADA, KOLHAPUR

A
TTL Project Report
On

"Secure Student Document System"

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Class: SY (CSE)

Div.: A
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CERTIFICATE

This is to certify that **Snehal Tanaji Khot** has successfully completed the term work for **TTL Project** entitled “**Secure Student Document System**”, towards the partial fulfillment of **S.Y. (CSE) Sem II** during the academic year **2025-26**.

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2025-26**

DECLARATION

I hereby declare that the dissertation work report entitled “**Secure Student Document System**” which is being submitted to **D.Y. Patil College of Engineering and Technology, Kolhapur**, in partial fulfillment of SY CSE TTL Project course, is a bonafide report of the work carried out by me. The material contained in this report has not been submitted to any university or institution for the award of any degree.

Place: Kolhapur.

Date:

Student Name

Snehal Tanaji Khot

Guide Name

Mrs. Rutuja Shivraj Kalekar

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Sincerely,

Student Name

Snehal Tanaji Khot

SY CSE TTL Project

INDEX

	Page No
Chapter 1: INRODUCTION	
1.1 Introduction	1
1.2 Technology	2
1.3 Motivation	3
Chapter-2: PROBLEM STATEMENT	
2.1 Need of Work	4
2.2 Problem Statement	4
2.3 Objectives	5
Chapter 3: DESIGN DETAILS	
3.1 System Architecture	6
3.2 System Design Diagrams	7
Chapter 4: IMPLEMENTATION	
4.1 Module Information	8
Chapter 5: EXPERIMENTAL RESULTS	
5.1 Screenshots	11
Chapter 6: CONCLUSION	
6.1 Conclusion	14
6.2 Future Scope	15
Chapter 7: REFERENCES	16

Chapter-1

INTRODUCTION

1.1 Introduction:

In today's digital era, educational institutions and organizations are increasingly shifting toward online systems for storing and managing student records. Important documents such as mark sheets, certificates, identity proofs, transfer certificates, and academic records are now commonly shared and maintained in digital form. Although this digital transformation has improved accessibility and reduced paperwork, it has also created serious challenges related to document security, privacy, and authenticity.

Many traditional systems used for document management are not secure enough to protect sensitive student information. Documents stored in ordinary databases can be easily modified, duplicated, or accessed by unauthorized users. In many cases, institutions still depend on manual verification methods, which are time-consuming, inefficient, and prone to human errors. As a result, there is always a risk of fake certificates, data tampering, and misuse of student records.

To overcome these issues, the Student Document Security System is developed as a secure web-based platform that uses modern technologies such as hashing and blockchain. The system focuses on providing strong security, transparency, and reliability in document storage and verification.

The project allows students to upload and manage their important documents securely. Every uploaded document is assigned a unique SHA-256 hash value, which acts as a digital fingerprint of that file. This hash value is then stored on the Ethereum blockchain through smart contracts. Since blockchain data cannot be modified once stored, it ensures that the originality and integrity of documents are always maintained.

The system also provides secure login authentication using Firebase Authentication. Users can log in safely using their email and password, and they receive login notifications whenever account access is detected. This improves security and helps users identify unauthorized login attempts.

Another important feature of the system is document verification. Whenever a document needs to be verified, the system generates a new hash value and compares it with the hash stored on the blockchain. If both values match, the document is considered original and untampered. If the values do not match, the system identifies the document as modified.

Overall, the Student Document Security System provides a reliable and efficient solution for secure document management. It reduces manual verification work, increases transparency, improves trust, and ensures the safety of important student records.

Student Document Security System

1.2 Technology Stack

Use of Trending Technology – Blockchain and Cryptography

Blockchain technology is one of the most advanced and secure technologies used in today's digital world. It provides decentralized, transparent, and tamper-proof data storage. In this project, blockchain is used to store document hash values securely so that no unauthorized modifications can occur.

Cryptography is another important technology used in this system. The SHA-256 hashing algorithm generates a unique digital identity for every uploaded document. Even a small change in the document creates a completely different hash value, making tampering detection easy and accurate.

The combination of blockchain and hashing technologies makes the system highly secure, transparent, and trustworthy.

1. Frontend (Client-Side)

- HTML5 – Used for creating the structure of web pages.
- CSS3 – Used for styling and responsive design.
- JavaScript – Used for implementing dynamic functionality and interactions.

2. Backend and Cloud Services

- Firebase Authentication – Provides secure user login and authentication.
- Firebase Database – Stores user-related information securely.
- Firebase Cloud Storage – Used for secure document storage.

3. Security Technologies

- SHA-256 Algorithm – Generates unique hash values for uploaded documents.
- Blockchain Technology – Stores document hashes in a secure and tamper-proof way.
- Ethereum Blockchain – Decentralized blockchain platform used for storage.
- Solidity – Programming language used to create smart contracts.

4. Development Tools

- Visual Studio Code – Code editor used for development.
- MetaMask Wallet – Connects the system with Ethereum blockchain.
- Ganache – Local blockchain testing environment.
- Web Browser – Chrome or Edge browser.

Student Document Security System

Summary Description for Report

“The Student Document Security System is developed using HTML, CSS, and JavaScript along with Firebase services for authentication and cloud storage. Advanced security mechanisms such as SHA-256 hashing and Ethereum blockchain are integrated to provide secure document storage, authentication, and tamper-proof verification.”

1.3 Motivation

The motivation behind this project comes from the increasing need for secure management of student documents in digital platforms. Educational institutions are rapidly adopting online systems, but many existing systems do not provide proper protection against unauthorized access, document tampering, and data misuse.

Students frequently need to share their certificates, mark sheets, and identity documents with colleges, companies, and organizations. However, traditional systems do not guarantee document authenticity and security. Manual verification processes consume a lot of time and are prone to errors.

This project aims to solve these issues by creating a secure digital system using blockchain and cryptography. The use of blockchain ensures transparency and immutability because once data is stored on the blockchain, it cannot be modified. The SHA-256 hashing mechanism provides document integrity and detects tampering.

Another important motivation is to improve trust and reliability in document verification systems. By automating verification and ensuring secure storage, the project reduces manual effort and increases efficiency.

Ultimately, the Student Document Security System aims to provide a secure, transparent, reliable, and user-friendly platform for managing important student documents.

Chapter-2

PROBLEM STATEMENT

2.1 Need of Work:

- Increasing use of digital document systems in educational institutions
- Risk of unauthorized access and data breaches
- Need for secure storage of student records
- Lack of reliable document verification systems
- Manual verification processes are slow and inefficient
- Requirement for tamper-proof document management
- Need to improve transparency and trust
- Requirement for secure online document sharing
- Need to reduce fake certificates and document fraud
- Demand for automated and efficient verification systems

2.2 Problem Statement

In many educational institutions and organizations, student documents are still managed using traditional systems that do not provide strong security mechanisms. Sensitive documents such as certificates, mark sheets, and ID proofs are vulnerable to unauthorized access, modification, duplication, and misuse.

Most existing systems rely on centralized databases that can be hacked or manipulated easily. Users are often unaware of suspicious login activities, increasing the chances of account hacking. There is also no reliable method to verify whether a document has been modified after uploading.

The document verification process is mostly manual, which consumes time and increases human errors. Institutions spend significant effort verifying the authenticity of documents. Furthermore, the lack of transparency in document handling reduces trust in digital systems.

Therefore, there is a strong need for a secure, transparent, and efficient system that can safely store, manage, and verify student documents using modern technologies such as blockchain and cryptography.

2.3 Objectives:

- To develop a secure web-based document management system

Student Document Security System

- To provide secure user login using email and password
- To send login notifications for security purposes
- To allow secure document upload and storage
- To generate SHA-256 hash values for uploaded documents
- To store document hashes securely on blockchain
- To provide fast and reliable document verification
- To detect document tampering effectively
- To reduce manual verification efforts
- To improve transparency and trust in document management

DESIGN DETAILS

1.1 System Architecture:

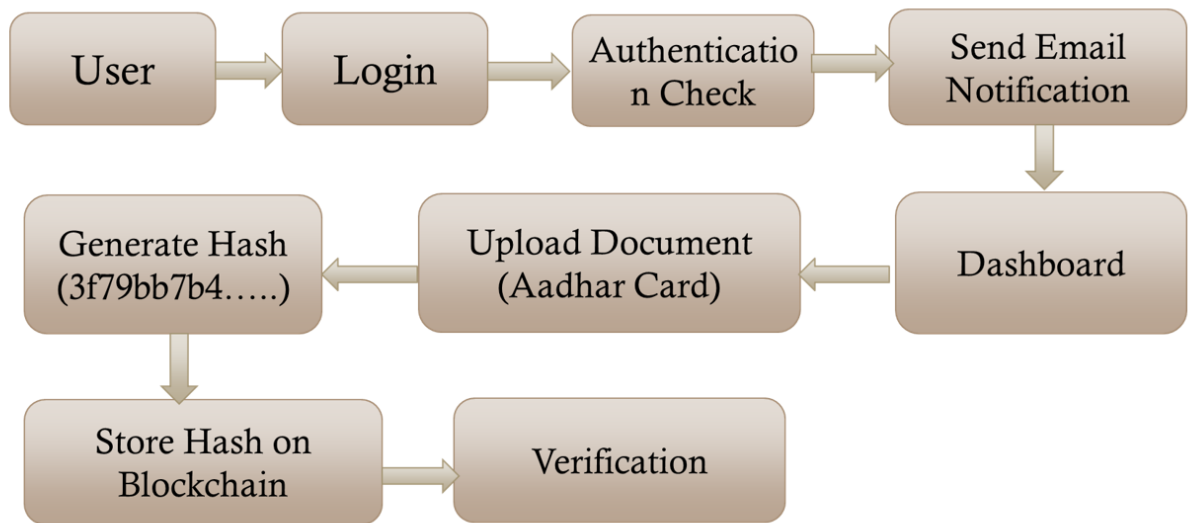


Fig.1 System Architecture

3.2 System Design Diagrams

Student Document Security System

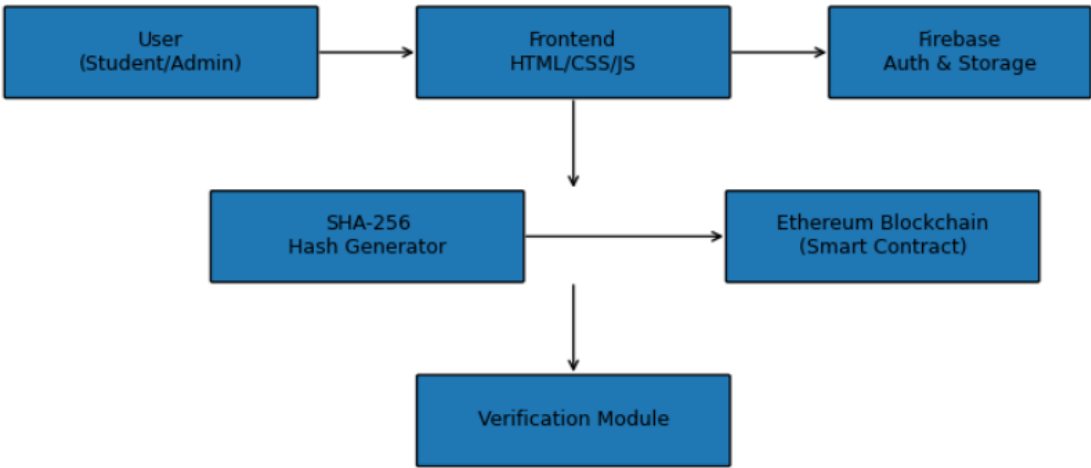


Fig.3.2.1 data flow Diagram

3.2.2Sequence Diagram

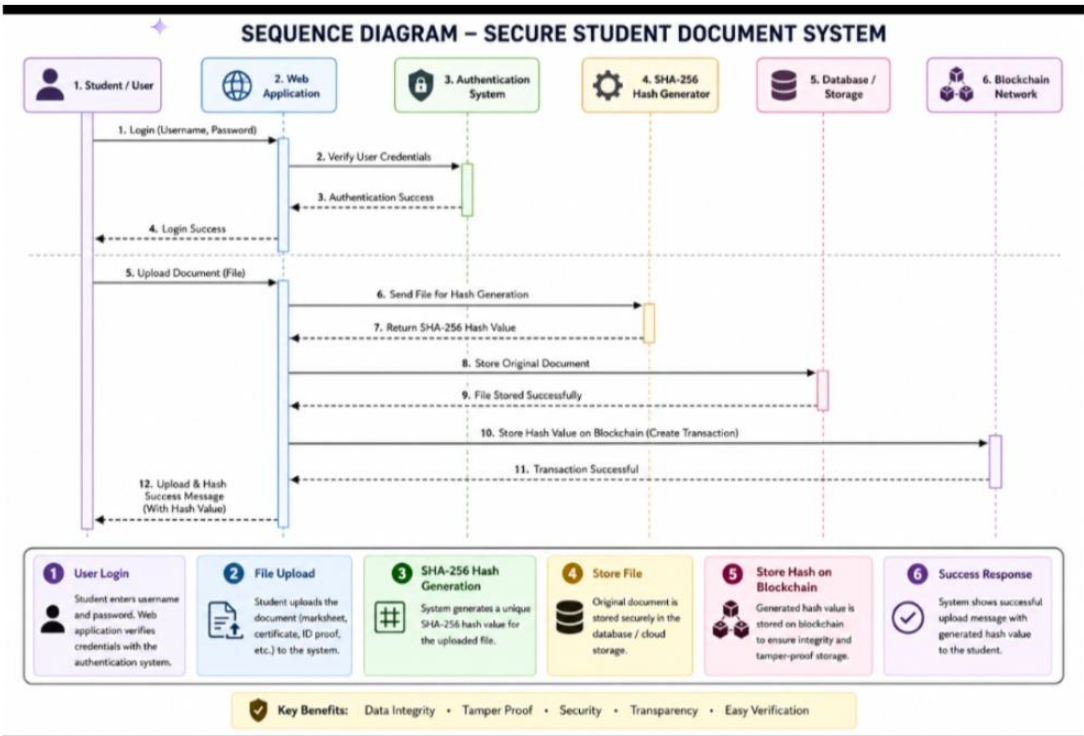


Fig.3.2.2 Sequence Diagram

Chapter – 4

IMPLEMENTATION

4.1 Module Description In detail

This chapter explains how the system is actually built and how each part works together. The Student Document Security System is developed as a web-based application that focuses on security, ease of use, and reliability. The system is divided into different modules, where each module performs a specific function. All modules are connected to provide a smooth and secure experience to the user.

4.1 Module Description :

1. Authentication Module

This is the first and most important part of the system. It allows users to log in using their email and password. Firebase authentication is used to manage user accounts securely. Whenever a user tries to log in, the system verifies the credentials and grants access only if the details are correct.

An important feature of this module is the **login notification system**. Whenever a login attempt is made, the user receives an email alert. This helps in identifying unauthorized access and increases overall security. This module ensures that only authorized users can access their documents.

2. Document Upload Module :

This module allows students to upload their important documents such as certificates, mark sheets, and ID proofs. The uploaded documents are stored securely in cloud storage using Firebase.

The system is designed in a way that uploading is simple and user-friendly. Users can easily select and upload files without any technical difficulty. Once uploaded, the documents are stored safely and can be accessed anytime by the user.

3. Hash Generation Module :

After a document is uploaded, the system generates a unique hash value using the SHA-256 algorithm. This hash acts like a digital fingerprint of the document.

Even a small change in the document will result in a completely different hash value. This helps in detecting any kind of modification or tampering. The hash value ensures that the originality of the document is maintained and can be verified at any time.

4. Blockchain Module :

The generated hash value is stored in the blockchain using Ethereum technology. Blockchain is used because it provides a secure and tamper-proof way of storing data.

Once the hash is stored in the blockchain, it cannot be changed or deleted. This ensures that the document remains secure forever. Solidity is used to write the smart contracts that manage the storage of hash values on the blockchain. This module plays a key role in maintaining transparency and trust in the system.

5. Verification Module :

This module is used to check whether a document is original or has been modified. When a document is verified, the system generates a new hash value and compares it with the hash stored in the blockchain.

If both hash values match, the document is considered valid and unchanged. If they do not match, it means the document has been tampered with. This process is fast and accurate, making document verification simple and reliable.

Student Document Security System

4.2 System Requirements

Software Requirements :

- **HTML, CSS, JavaScript:**
These are used to design and develop the frontend of the system. They help in creating a simple and interactive user interface.
 - **Firebase:**
Firebase is used for authentication, database management, and cloud storage. It helps in securely storing user data and documents.
 - **Ethereum Blockchain:**
This is used to store the hash values of documents securely. It ensures that the data cannot be modified.
 - **Solidity:**
Solidity is used to write smart contracts that interact with the blockchain.
 - **Web Browser:**
Any modern browser like Chrome or Edge is required to run the system.
-

Hardware Requirements :

- **Computer or Laptop:**
A basic system with standard configuration is enough to run the application.
 - **Internet Connection:**
A stable internet connection is required to access Firebase services and blockchain network.
-

Development Approach : The system can be developed using a simple step-by-step approach. First, the frontend is created using HTML, CSS, and JavaScript. Then Firebase is integrated for login and storage. After that, the SHA-256 hashing function is implemented. Finally, blockchain integration is done to store hash values securely.

Chapter -5

EXPERIMENTAL RESULTS

5.1 Experimentation:(Output Screenshots with description)

This chapter explains how the system was tested and what results were obtained after implementation. The Student Document Security System was tested step by step to check whether all modules are working properly. The testing was done in a real-time environment using a web browser, Firebase services, and blockchain integration.

The main aim of testing was to ensure that the system is secure, easy to use, and performs all operations correctly such as login, document upload, hash generation, and verification.

To test the system, different operations were performed by the user. Each function of the system was checked carefully to confirm that it works as expected.

1. User Login Testing

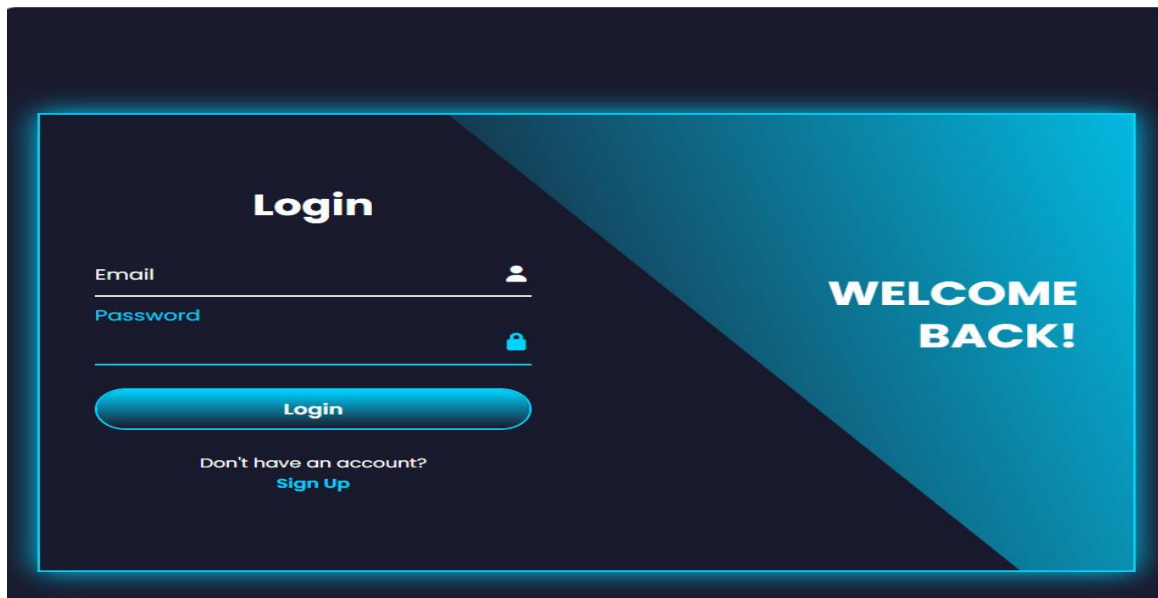
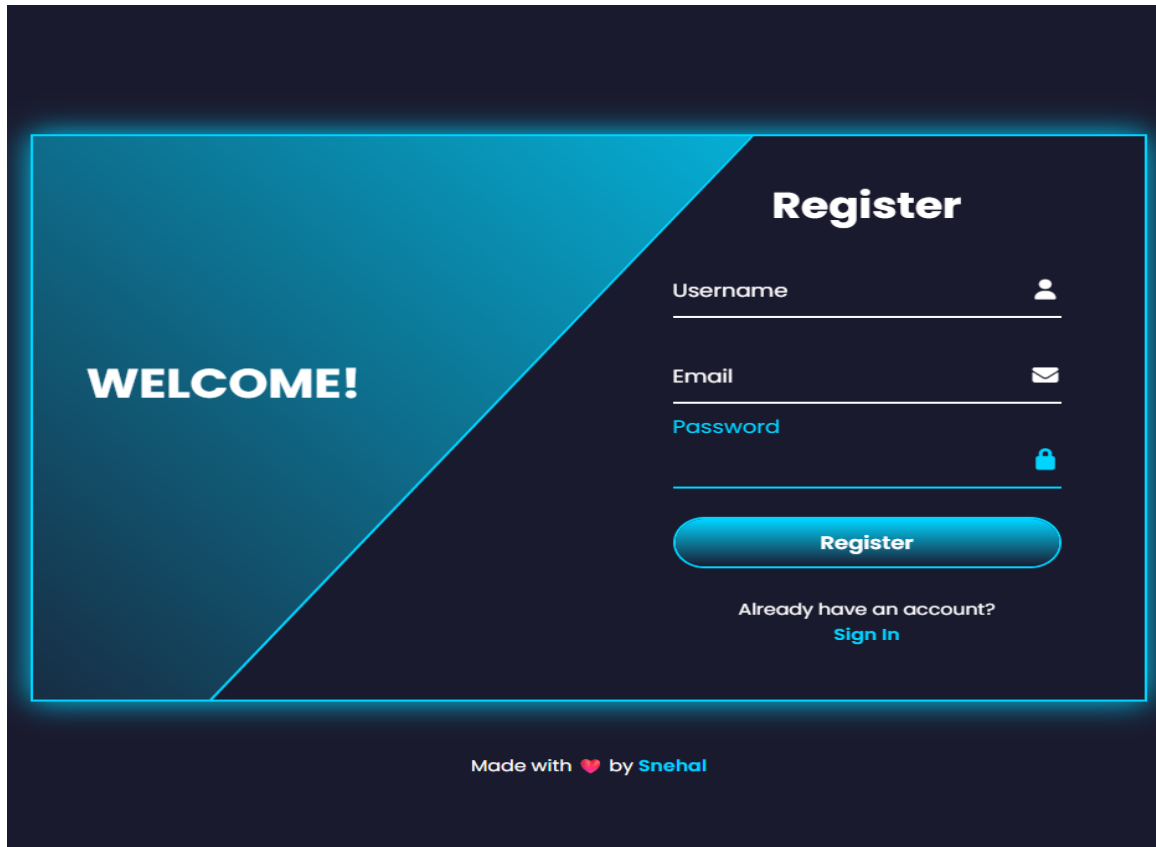
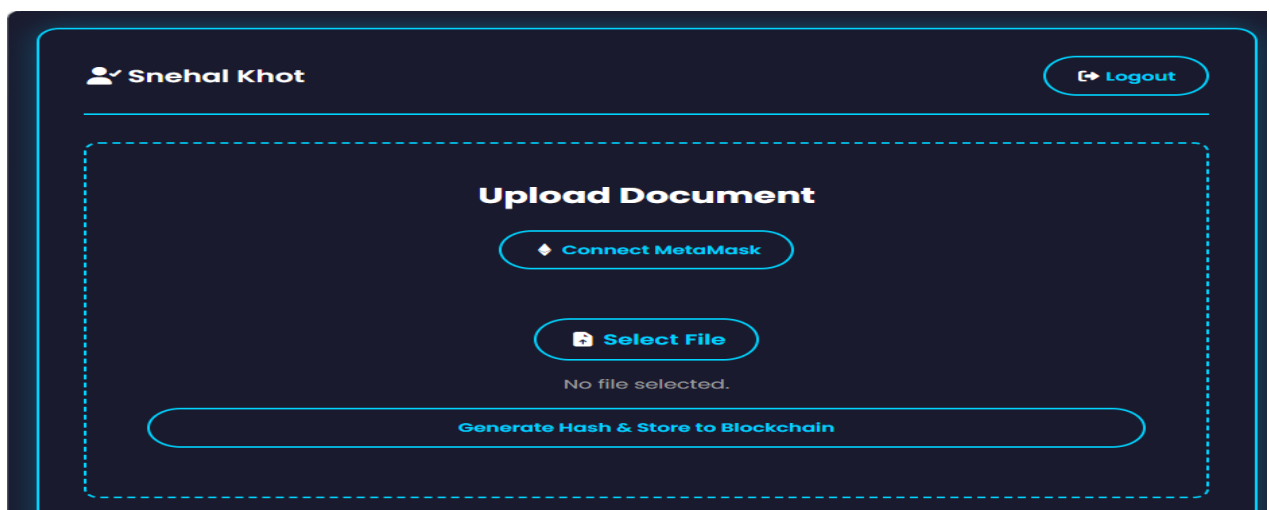


Fig 1: Login page First, the user logged into the system using email and password. The system successfully authenticated the user using Firebase. After login, an email notification was sent to the user. This confirmed that the login alert feature is working properly.



The image shows a 'Register' page with a dark blue background. On the left, a teal diagonal banner contains the word 'WELCOME!' in white. On the right, the 'Register' title is in white. Below it are three input fields: 'Username' with a person icon, 'Email' with an envelope icon, and 'Password' with a lock icon. A teal 'Register' button is below the fields. Under the button, the text 'Already have an account?' is followed by a teal 'Sign In' link. At the bottom center, it says 'Made with ❤️ by Snehal'.

Fig 2: sign up page First, the user logged into the system using email and password. The system successfully authenticated the user using Firebase. After login, an email notification was sent to the user. This confirmed that the login alert feature is working properly.



The image shows a user dashboard with a dark blue background. At the top left, a user profile icon is next to the name 'Snehal Khot'. At the top right, there is a teal 'Logout' button. The main content area is enclosed in a dashed teal border and titled 'Upload Document' in white. Inside this area, there are two buttons: 'Connect MetaMask' with a diamond icon and 'Select File' with a document icon. Below these buttons, the text 'No file selected.' is displayed. At the bottom of the dashed area is a teal button labeled 'Generate Hash & Store to Blockchain'.

Student Document Security System

Fig 3: uploading documents After login, the user uploaded a document such as a certificate or mark sheet. The system accepted the file and stored it securely in the cloud storage. The upload process was smooth and user-friendly.

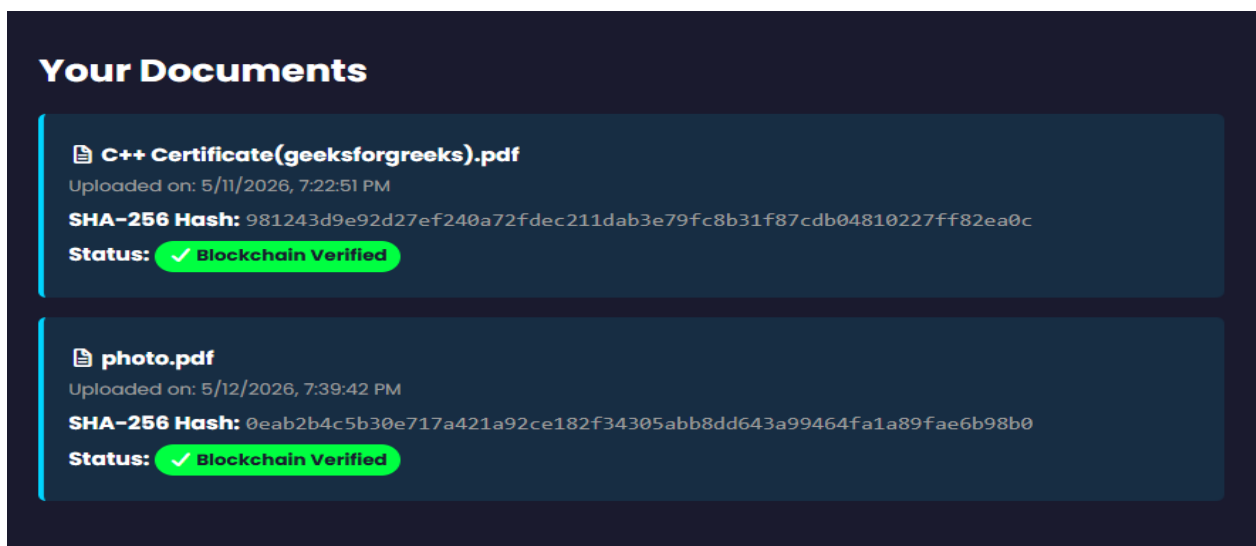


Fig 5: Hash Value : Once the document was uploaded, the system generated a SHA-256 hash value for that document. This hash was unique and confirmed that the document has its own digital identity.

4. Blockchain Storage Testing

The generated hash value was stored in the blockchain. The storage process was successful, and the hash could not be modified after storing. This ensured that the document is protected from tampering.

Results

The results of the system testing are as follows:

- The login system worked correctly with proper authentication
- Email notifications were successfully sent on login attempts
- Documents were uploaded and stored securely
- Unique hash values were generated for each document
- Blockchain stored the hash values safely

Overall, the system performed all operations correctly and provided a secure environment for managing student documents.

CONCLUSION

6.1 Conclusion:

The Student Document Security System was successfully designed and developed to provide secure storage and verification of important student documents. The project effectively addresses major issues such as unauthorized access, document tampering, fake certificates, and inefficient manual verification systems.

The system uses modern technologies such as Firebase Authentication, SHA-256 hashing, and Ethereum blockchain to provide strong security and transparency. Firebase ensures secure login and user management, while SHA-256 generates unique digital fingerprints for uploaded documents. Blockchain technology guarantees that document hash values remain immutable and tamper-proof.

One of the major achievements of this system is the implementation of automated document verification. Instead of depending on manual checking methods, the system verifies document authenticity quickly and accurately by comparing hash values. This reduces human effort, saves time, and improves reliability.

The project also provides login notifications that help users detect suspicious account activity. The user-friendly interface makes the system simple and accessible for students and institutions.

Overall, the Student Document Security System successfully fulfills its objectives by providing:

- Secure document storage
- Tamper-proof verification
- Fast authentication
- Transparency and reliability
- Reduced manual verification work

The system proves to be an effective solution for modern digital document management.

6.2 Future Scope

Although the system works effectively, several advanced features can be added in the future to make it more powerful and useful.

1. QR Code Verification

Documents can be linked with QR codes for quick and easy verification.

Student Document Security System

2. Mobile Application Development

A mobile application can be developed so users can access documents anytime and anywhere.

3. Multi-University Integration

The platform can support multiple colleges and universities on a single system.

4. Two-Factor Authentication

Additional security methods such as OTP verification can improve account protection.

5. AI-Based Document Management

Artificial Intelligence can automatically organize and categorize uploaded documents.

6. Government Portal Integration

The system can be integrated with official educational and government portals for direct verification.

7. Multi-Language Support

Support for multiple languages can help users from different regions use the system easily.

8. Advanced Analytics Dashboard

Interactive dashboards and analytics can help institutions monitor document activities efficiently.

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2. Mastering Ethereum – Andreas M. Antonopoulos & Gavin Wood
3. Cryptography and Network Security – William Stallings
4. Software Engineering – Ian Sommerville
5. Web Technologies: HTML, JavaScript, PHP, Java – Uttam K. Roy
6. Learning Firebase – Greg Lim
7. JavaScript: The Definitive Guide – David Flanagan
8. Beginning Blockchain – Bikramaditya Singhal